Ekkati Laxman Reddy

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Summary

B.Tech Computer Science graduate focused on Generative AI, RAG pipelines, NLP, vector search, and LLM-powered applications. Built AI systems using Python, Streamlit, ChromaDB, Hugging Face, Groq API, BM25, reranking, and graph-based retrieval. Completed 5 months of Cognizant training with exposure to Java automation frameworks, Agile delivery, Git/GitHub, Jenkins CI/CD, and Selenium TestNG.

Skills

Languages: Python, SQL, Java
Generative AI & NLP: LLMs, Generative AI, RAG, Graph RAG, NLP, Transformers, Prompt Engineering, Semantic Search, Vector Search, Hugging Face
Frameworks & Tools: Streamlit, ChromaDB, FAISS, LangChain, LangGraph, FastAPI, Groq API, OpenAI API, Git, GitHub
Data Science: Pandas, NumPy, EDA, Data Preprocessing, Feature Engineering, Model Evaluation, Data Visualization
Testing & Automation: Selenium WebDriver, TestNG, Maven, Page Object Model, WebDriverWait, BDD Cucumber

Projects

DataGroq – AI Document Assistant Sept 2025

Python, Groq API, LLaMA-3, NLP, Machine Learning, Plotly, Streamlit

- Built an AI-powered document assistant enabling conversational Q&A over 200+ documents using Groq API and LLaMA-3.
- Developed an NLP pipeline for tokenization, stopword removal, lemmatization, document classification, and keyword extraction, improving analytical accuracy by 30% over baseline keyword search.
- Reduced manual document analysis effort by 70% through automated summarization, question answering, and visual analytics workflows.

Multi-Phase AI Retrieval and Analytics Feb 2026 – Mar 2026

Python, NLP, LLMs, RAG, Graph RAG, ChromaDB, Groq API, BM25, NetworkX, Streamlit

- Built a multi-phase RAG system over 500 arXiv research papers using dense retrieval, hybrid retrieval, and Graph RAG to support citation-aware question answering and research analytics.
- Combined BM25 sparse retrieval with bi-encoder dense retrieval using Reciprocal Rank Fusion and cross-encoder reranking, achieving Recall@5 of 0.76 and 43% MRR improvement over a single-retriever baseline.
- Designed a Graph RAG layer with NetworkX using BFS-based 1-hop citation expansion to retrieve related paper chunks and improve multi-document reasoning.

Traffic Analysis Dashboard Jan 2025 – Feb 2025

Tableau, Python, Google Sheets, Geoapify API

- Designed a real-time traffic analytics dashboard in Tableau using Google Sheets and Geoapify API data to visualize route efficiency, commercial hotspots, and traffic KPIs.
- Built an automated data extraction and transformation workflow across APIs, reducing manual processing effort by 80%.
- Created geospatial visualizations and KPI views to identify high-traffic zones, route patterns, and location-based operational insights for faster decision-making.

Work Experience

Programmer Analyst Trainee Jan 2026 – May 2026

Cognizant Technology Solutions

- Built and maintained a scalable Selenium TestNG automation framework using Java, Page Object Model, WebDriverWait, exception handling, and assertion-based validation.
- Automated end-to-end functional and UI testing workflows by developing reusable test components, improving coverage and reducing manual testing effort.
- Worked with Java, Selenium WebDriver, TestNG, Maven, Git/GitHub, Jenkins CI/CD, and BDD Cucumber in an Agile environment, collaborating with senior engineers on reliable software delivery.

Education

Lovely Professional University Phagwara, Punjab
Bachelor of Technology, Computer Science and Engineering – CGPA: 8.1 *Completed July 2026*

Sri Chaitanya Junior College Hyderabad, Telangana
Intermediate – 96.8% *2021 – 2022*

Certifications & Achievements

AWS Certified Generative AI Developer – Credly Apr 2026
Google Cloud Associate Data Practitioner – Credly Mar 2026
Data Visualization with Tableau – Coursera Nov 2024
5-Star Rating in SQL – HackerRank Feb 2025